Bangladesh Army University of Engineering & Technology (BAUET)

Qadirabad Cantonment, Natore-6431

Lesson Plan

1st Year 1st semester

Course Title: Chemistry

Credit Hour: 2.00

Course Instructor: Dr. Tasmina Khandaker

Lecturer, Department of Chemistry

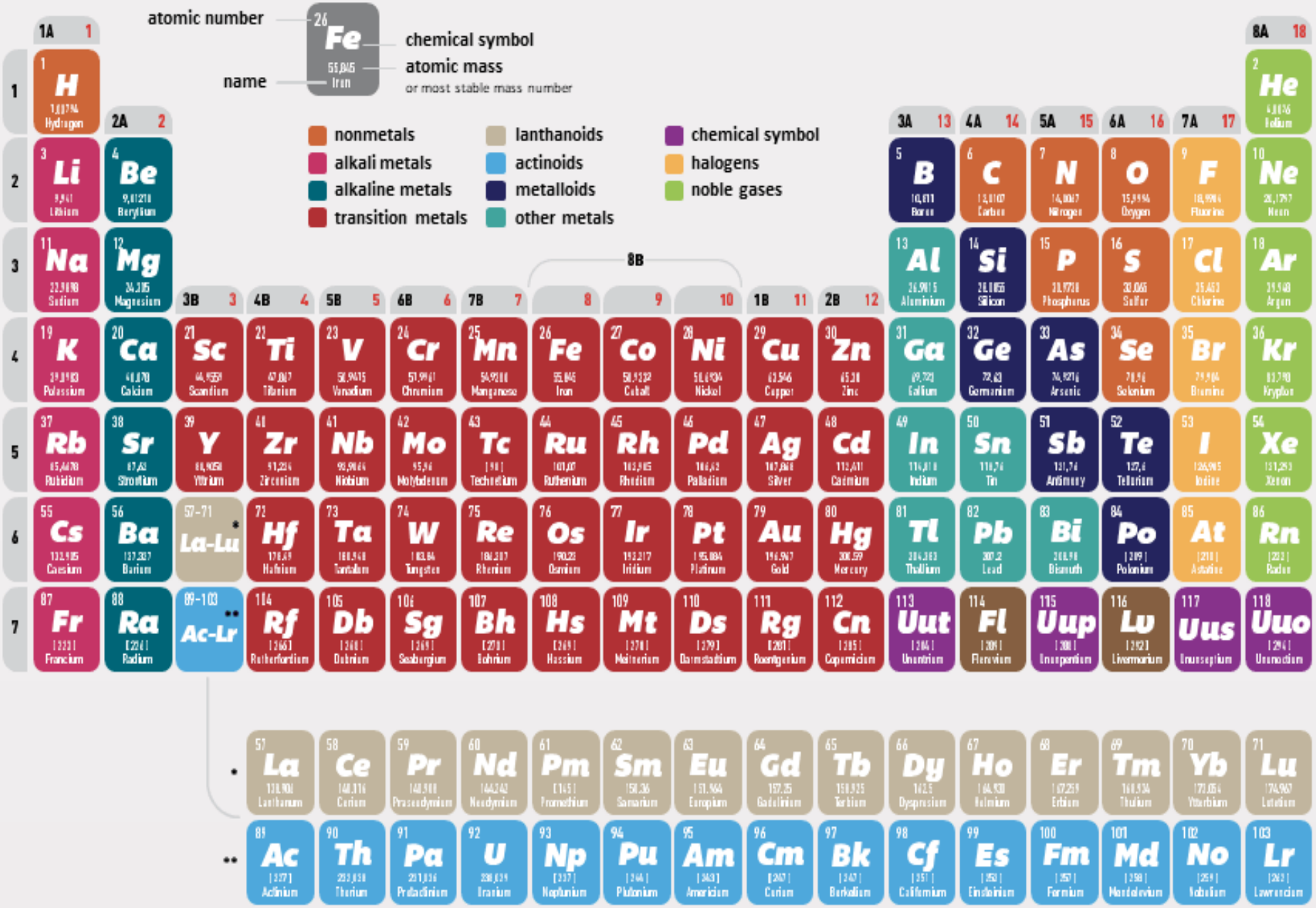
Reference Books:

1. Introduction to Modern Inorganic Chemistry- S. Z. Haider
2. Modern Inorganic Chemistry- R. D. Madan
3. Fundamental Concepts of Inorganic Chemistry- S. Gilreath:
4. Advanced Inorganic Chemistry- S. Prakash & G.D Tuli
5. Essentials of Physical Chemistry- B. S. Bahl and G. D. Tuli
6. Principle of Physical chemistry- M. M. Haque and M. A. Nawab
7. A text book of Physical Chemistry- S Glasstone
8. Essentials of Physical Chemistry- B. S. Bahl and G. D. Tuli

Distribution of Marks:

Class Participation	5%
Observations	5%
Homework/ Assignment	5%
Class Test/Quiz	15%
Final Examination	70%
Total	100%

Modern Periodic Table



Modern Periodic Law

- The modern periodic law states that the physical and chemical properties of the elements are the periodic functions of their atomic numbers.
- Scientists arranged elements in increasing order of their atomic numbers from left to right across each row. And discovered that the elements having similar properties repeat after regular intervals.

Why does the modern periodic table depend on atomic number rather than atomic mass?

- Atomic mass is the total mass of the protons and neutrons present in a nucleus of an atom. Whereas, the atomic number is the number of protons in a nucleus which determine the elements chemical properties. Also, the number of protons in the nucleus is equal to the electrons present outside the nucleus.
- The concept of arranging elements by atomic number was proposed by Henry Moseley in 1913. Prior to Moseley's work, elements were primarily arranged by atomic mass. However, the discovery of isotopes—elements with the same number of protons but different numbers of neutrons—highlighted the limitations of organizing elements solely by atomic mass. However, isotopes of the same element share similar chemical properties because they have the same number of protons (same atomic number).
- We know that **the nucleus is deep-seated inside an atom**. **But the electrons outside it, especially the ones in the outermost shell, are free to move around**. Hence they take part in chemical reactions. For this reason, the properties of an element depend on the atomic number rather than the atomic mass.

What is the basic difference between the Mendeleev's and the Modern Periodic Law?

Mendeleev's Periodic Law:

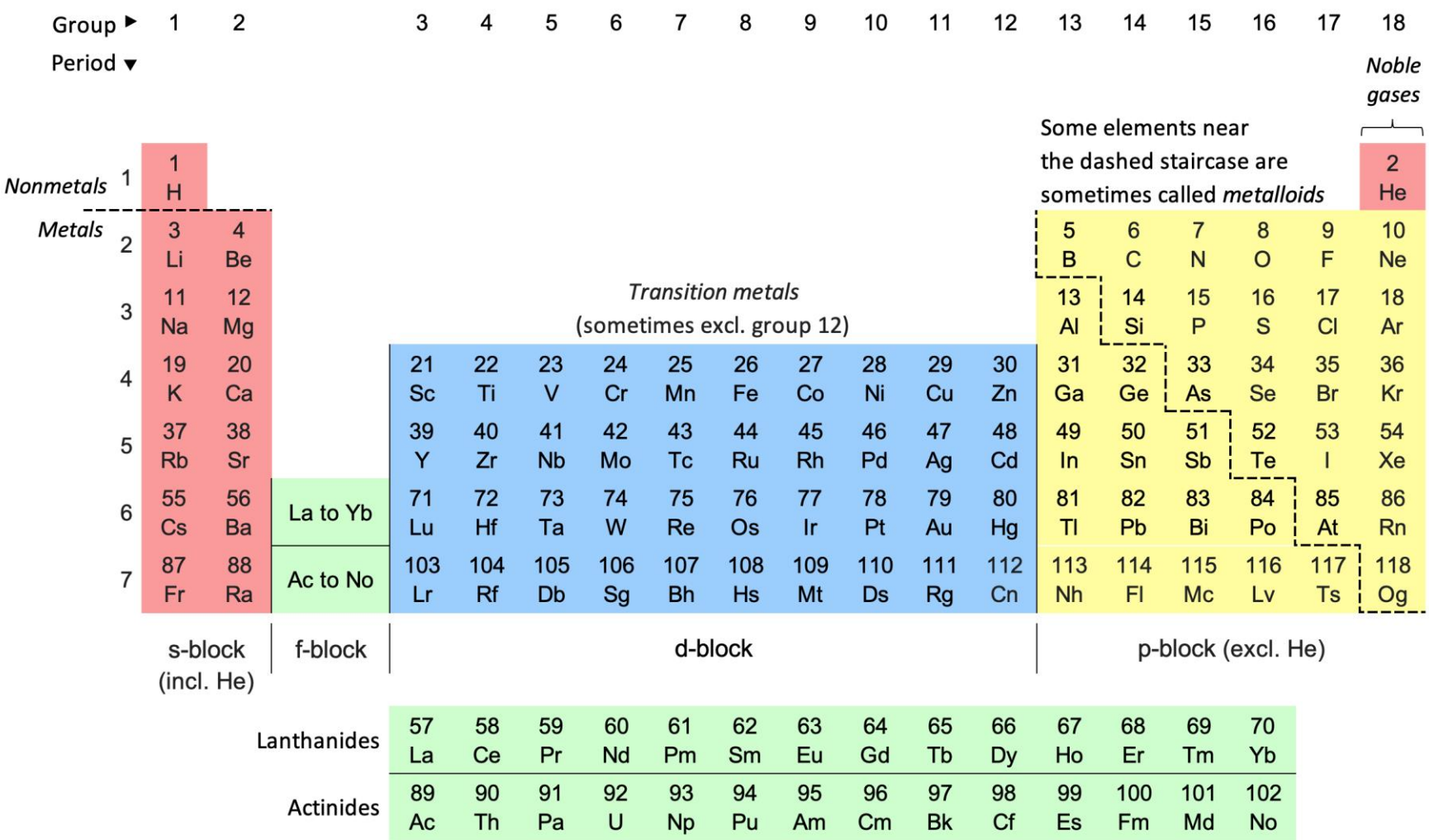
- Mendeleev's periodic law was formulated in the late 19th century.
- Mendeleev organized the elements primarily based on their atomic masses and chemical properties.
- He arranged the elements in increasing order of atomic masses and grouped elements with similar properties into columns or groups.
- Mendeleev's periodic table left gaps for elements that were yet to be discovered, and he even predicted the properties of some of these missing elements based on the patterns observed in the table.

Modern Periodic Law:

- The modern periodic law is based on the atomic number of elements rather than atomic mass.
- It was developed in the early 20th century after the discovery of the atomic structure and the realization that atomic number (the number of protons in an atom's nucleus) is a more fundamental property than atomic mass.
- Elements are arranged in order of increasing atomic number, and their properties are periodic functions of atomic number.
- The modern periodic table also organizes elements into groups with similar chemical properties, but the organization is based on the number of valence electrons rather than atomic mass.

Long Form of Periodic Table

The modern or long form of the periodic table is based on the modern periodic law. The table is the arrangement of elements in increasing order of their atomic numbers. The modern periodic table is the present form of the periodic table. And **it consists of 18 vertical columns (group) and 7 horizontal rows (Period)**



Groups in the Modern Periodic Table

- Groups also known as families are the vertical columns in the modern or long form of the periodic table.
- There are 18 groups in the periodic table.
- Each group consists of elements having the same outer shell electronic configuration. That means elements have similar chemical properties due to their similarities in the number of electrons in their outer shells

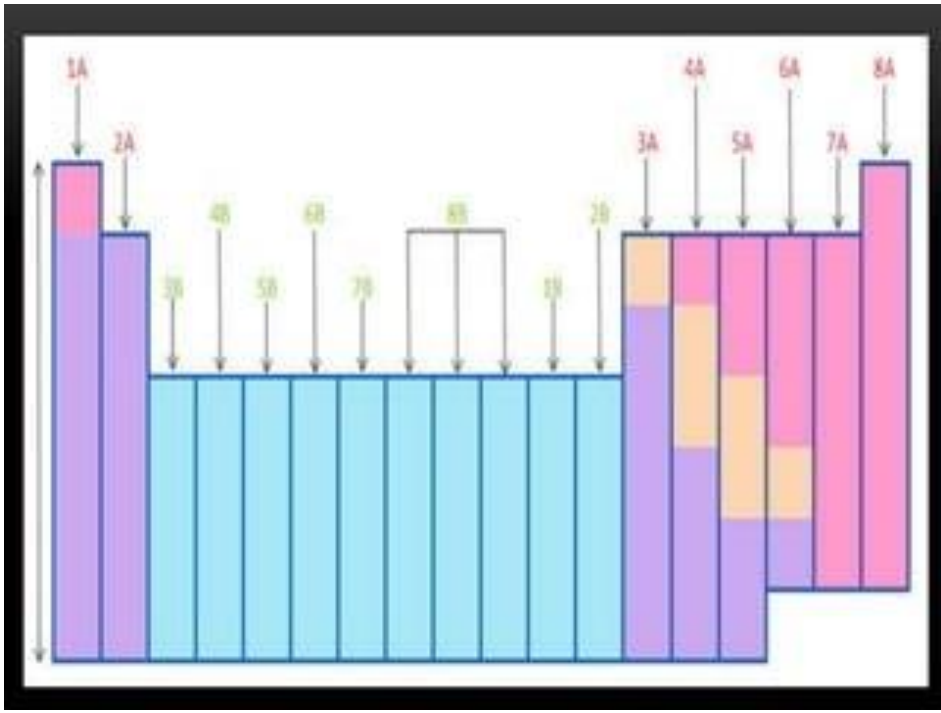
- **composed of two categories:**

a. Group A (Representative Elements)

b. Group B (Transition Metals)

GROUP A (REPRESENTATIVE ELEMENTS)

- Group IA – Alkali Metals (except Hydrogen)
- Group IIA – Alkaline Earth Metals
- Group IIIA – Aluminum Group/Boron Family
- Group IVA – Carbon Family
- Group VA – Nitrogen Family
- Group VIA – Oxygen Family/Chalcogens
- Group VIIA – Halogens
- Group VIIIA – Noble/Inert Gases



GROUP B (TRANSITION ELEMENTS)

- found in between Group IIA and Group IIIA and referred to as the d-block
- inner transition elements (lanthanides and actinides) are found at the bottom of the table and referred to as the f-block

Periods in the Modern Periodic Table

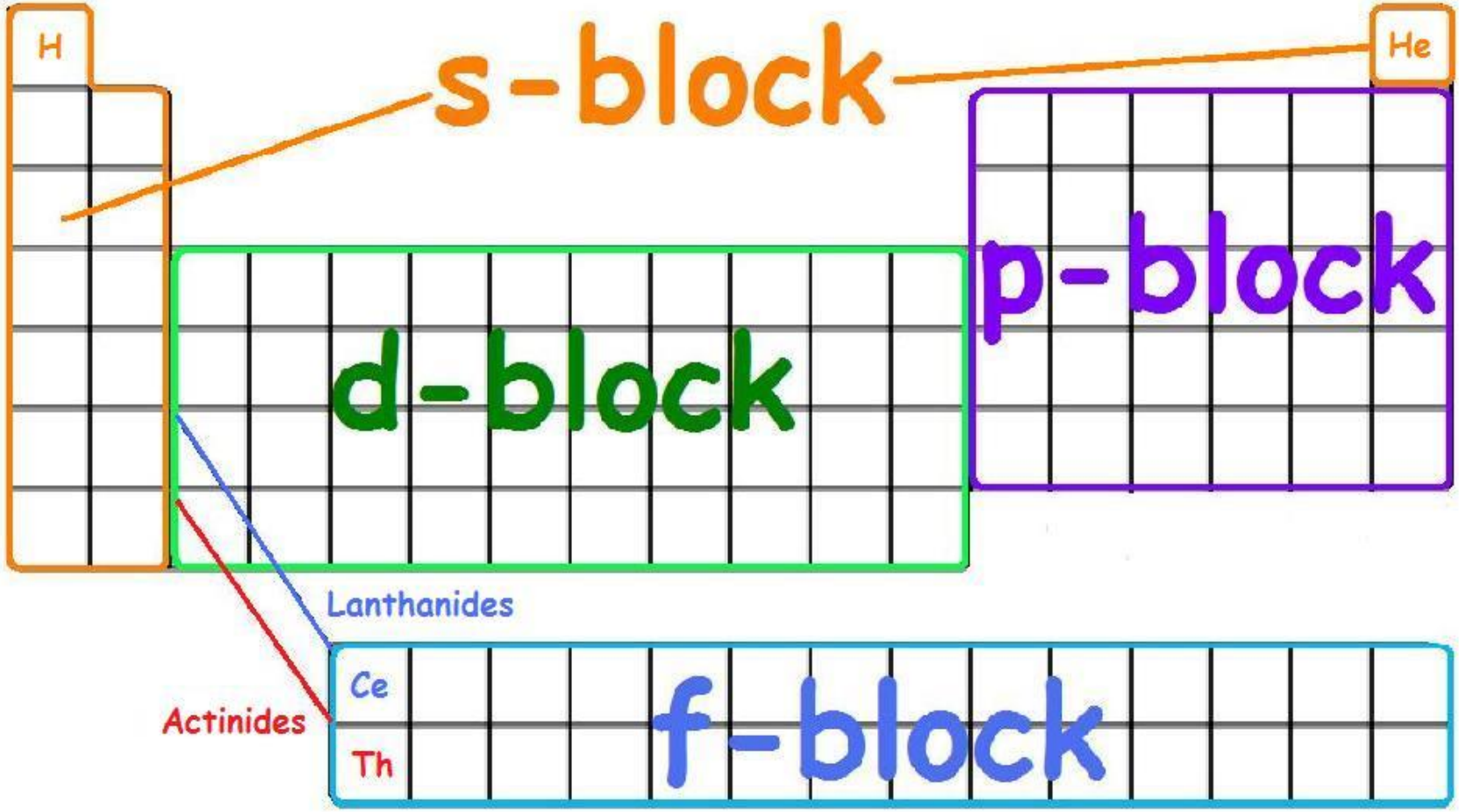
- Periods are the horizontal rows in the modern or long form of the periodic table.
- There are 7 periods in the periodic table.
- These are numbered as 1, 2, 3, 4, 5, 6 and 7 from top to bottom.
- The 1st period consists of only two elements – Hydrogen and Helium.
- While the 2nd and 3rd period consists of 8 elements each.
- The 4th and 5th period consists of 18 elements each
- On the other hand, the 6th period consists of 32 elements.
- The 7th period of the periodic table now has four new elements. They are 113-Nihonium, 115-Moscovium, 117-Tennessine, and 118 –Oganesson. This addition has completed the 7th period with 32 elements.
- Also, the long form of the periodic table consists of a separate panel at the bottom. It consists of 14 elements of the 6th period called the lanthanoids. And 14 elements in the 7th period called the actinoids.
- Each period represents the number of shells or energy levels present in an atom of an element.

Cause of Periodicity in the Modern Periodic Table

- The cause of periodicity in properties is the repetition of similar outer electronic configuration after certain regular intervals.
- For example, all the elements of group 1 i.e. alkali metals have a similar outer electronic configuration, ns^1. Here n refers to the Principal Quantum Number of the outermost shell.
- In the same way, the elements of group 17 have a similar outer electronic configuration, $ns^2\ np^5$. Hence have similar properties to one another. They are halogens.
- Similarly, the elements of group 18 have an outer electronic configuration of $ns^2\ np^6$. They have full filled orbitals. They comprise of non-reactive elements called Inert gases. Helium and Argon have electron configuration similar to the elements of this group. So they are together and known as Inert gases!
- Likewise, elements in a group have similar properties to those of all the other members of the same group. This is because they all have a similar electron configuration of the outer shell.

FOUR MAIN REGIONS IN THE PERIODIC TABLE

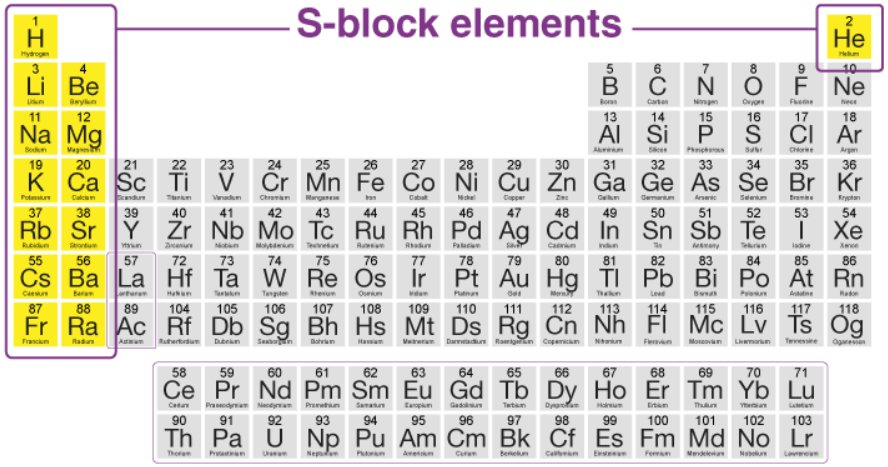
• Elements belonging to the same group have similar last sublevel configuration (lsc). Thus using the electron configuration of the elements, the periodic table can be divided into 4 regions or blocks such as s-, p-, d- and f-blocks.



Division of Elements into Blocks

s-block

The elements of Group 1 (alkali metals) and Group 2 (alkaline earth metals) which have ns^1 and ns^2 outermost electronic configuration belong to the s-Block Elements.



Characteristics of s-Block elements

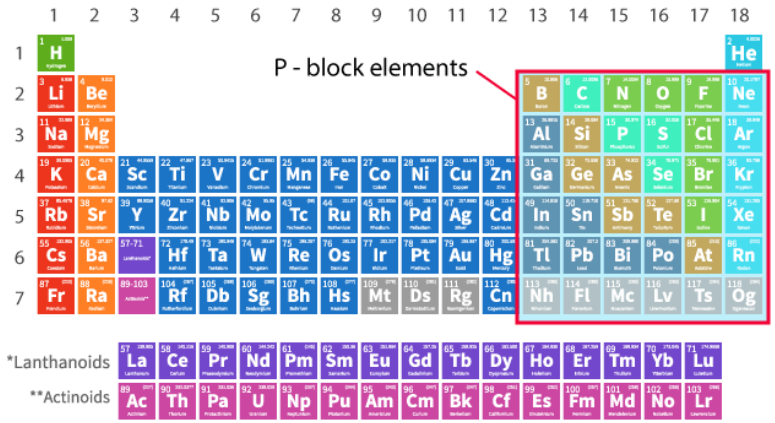
1. Except Be and Mg, all impart characteristic colour to the flame.
2. These have low ionization energy.
3. These are highly reactive.
4. These are the highly electropositive elements.
5. All the elements are soft metals.
6. They have low melting and boiling points.

Why does helium exhibit S-block characteristics?

Helium exhibits some characteristics similar to the elements in the s-block of the periodic table due to its electron configuration and valence shell structure.

- Helium has only two electrons, which occupy the 1s orbital. This makes its outermost electron shell similar to that of the s-block elements, which typically have one or two electrons in their outermost shell.
- Like the s-block elements, helium has a low electronegativity and tends to lose rather than gain electrons in chemical reactions. This is because it requires less energy for helium to lose its two electrons and achieve a stable electron configuration similar to that of a noble gas.
- Additionally, helium, like the s-block elements, has a relatively low atomic mass and density compared to elements found in other blocks of the periodic table

p-block: The p-Block Elements comprise those belonging to Group 13 to 18 and these together with the s-Block Elements are called the Representative Elements or Main Group Elements. The outermost electronic configuration varies from ns^2np^1 to ns^2np^6 in each period.



Characteristics of p-Block elements

1. The compounds of p-block elements are generally covalent although their ionic character increases down the group.

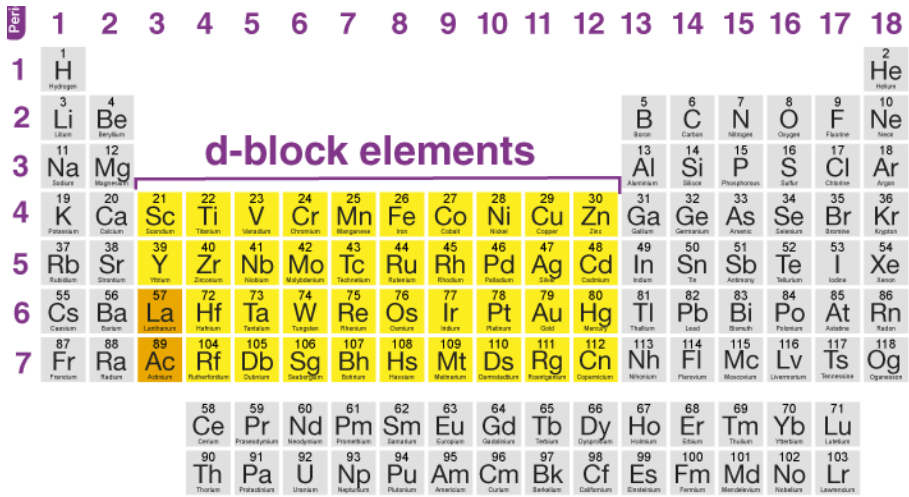
2. From left to right 13 to 18, reducing character decreases.

3. The p-block elements generally show more than one oxidation state.

4. The reactivity of elements in a group generally decreases downwards.

5. At the end of each period is a noble gas element with a closed valence shell $ns^2\ np^6$ configuration.

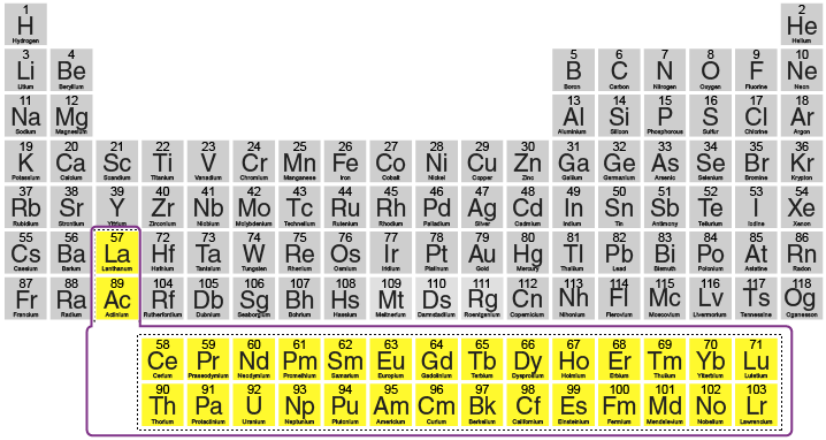
d-block: These are the elements of Group 3 to 12 in the center of the Periodic Table. These are characterized by the filling of inner d orbitals by electrons and are therefore referred to as d-Block Elements. These elements have the general outer electronic configuration $(n-1)d^{1-10}ns^{0-2}$.



Characteristics of d-Block elements

1. They are all metals with high melting and boiling points.
2. The compounds of the elements are generally paramagnetic in nature.
3. They mostly form colored ions, exhibit variable valence (oxidation states).
4. They are often used as catalysts.
5. These elements have high melting point.

f-block: The two rows of elements at the bottom of the Periodic Table, called the **Lanthanoids**, Ce(Z = 58) – Lu(Z = 71) and **Actinoids**, Th(Z=90) – Lr (Z=103) are characterised by the outer electronic configuration $(n-2)f^{1-14}(n-1)d^{0-1}ns^2$. The last electron added to each element is filled in f-orbital. These two series of elements are hence called the **Inner-Transition Elements** (f-Block Elements).



Characteristics of f-Block elements

- All actinoids are radioactive. Elements after uranium are known as trans-uranium element.
- They form coloured compounds.
- These two series of elements are called Inner Transition Elements
- They are all metals. Within each series, the properties of the elements are quite similar.
- They generally have high melting and boiling points.

Periodic Properties

The properties which generally have a regular trend along a group or period are called periodic properties. These are as given below :

1. Atomic size
2. Ionization energy
3. Electron gain enthalpy
4. Electronegativity

(i) Atomic Size

Atomic Radius is the distance from the center of the nucleus to the outermost shell containing electron.

Ionic Radius

The ionic radii can be estimated by measuring the distances between cations and anions in ionic crystals. In general, the ionic radii of elements exhibit the same trend as the atomic radii.

Cation: The removal of an electron from an atom results in the formation of a cation. The radius of cation is always smaller than that of the atom.

Anion: Gain of an electron leads to an anion. The radius of the anion is always larger than that ‘ of the atom.

(ii) Ionization energy

It is the amount of energy required to remove the outer most electron from an isolated atom in its gaseous state. It is the measured in the unit of kJ/mole. It is denoted by ($\Delta_i H$).

$$M(g) - e^- \longrightarrow M^+(g)$$

(iii) Electron gain enthalpy

It is the enthalpy change when an electron is added to the gaseous neutral atom. Electron gain enthalpy provides a measure of the ease with which an atom adds an electron to form anion. It is the measured in the unit of kJ/mole. It is denoted by ($\Delta_{eg} H$).

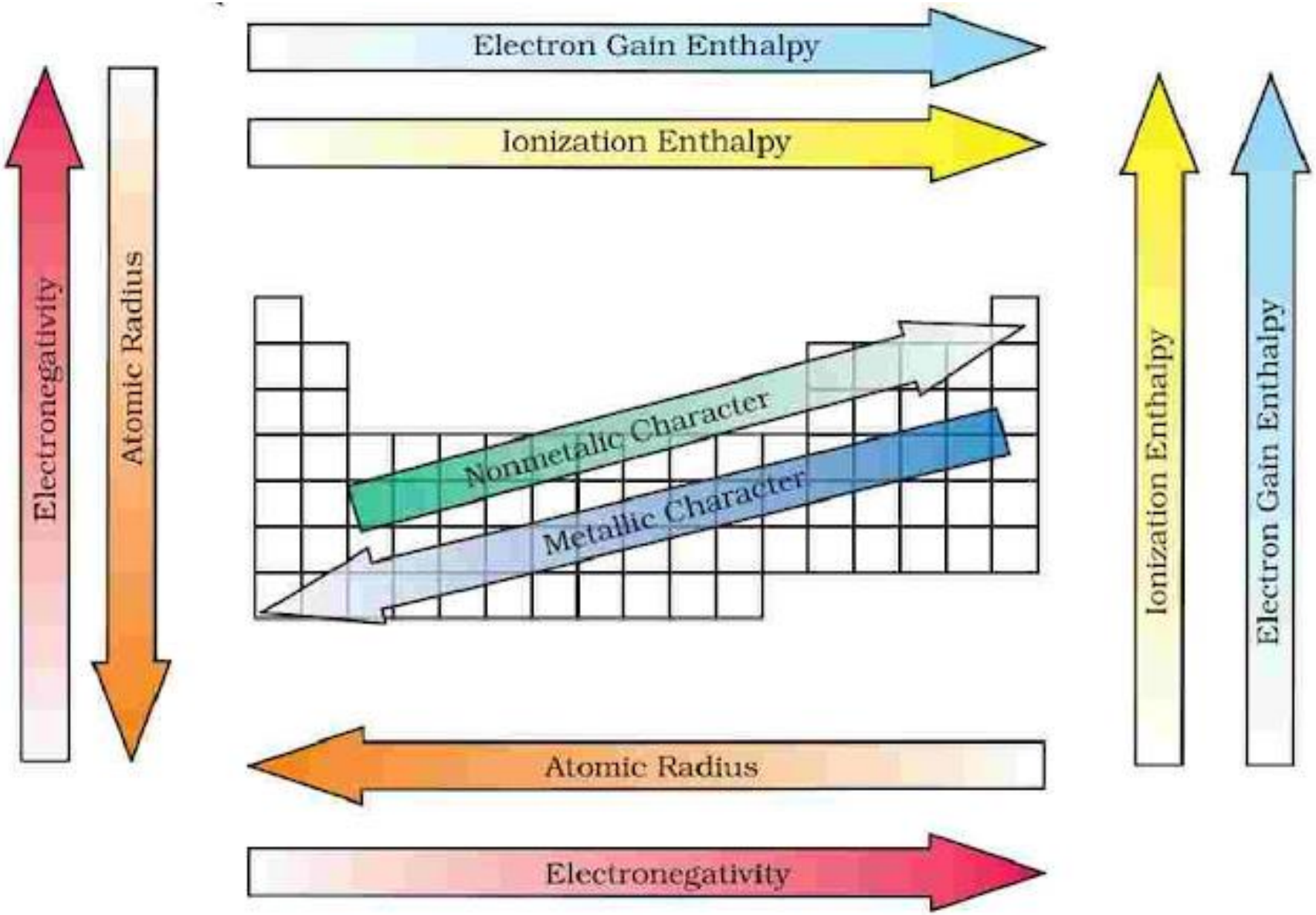
$$X(g) + e^- \longrightarrow X^-(g)$$

(iv) Electronegativity

Electronegativity is a measure of the tendency of an element to attract bonded electron pair towards itself in a covalently bonded molecule.

Periodic Trends in Chemical Properties along a Period

- **Metallic character**: It decreases across a period, maximum on the extreme left (alkali metals).
- **Non-metallic character**: It increases along a period, from left to right.
- **Atomic Size**: It decreases across a period.
- **Ionisation energy**: It increases along a period.
- **Electron gain enthalpy**: It increases along a period.
- **Electronegativity**: It increases along a period.
- **Basic nature of oxides**: It decreases from left to right in a period.
- **Acidic nature of oxides**: It increases from left to right in a period.


Electron Gain Enthalpy
Ionization Enthalpy
Electronegativity
Atomic Radius
Nonmetallic Character
Metallic Character
Ionization Enthalpy
Electron Gain Enthalpy
Atomic Radius
Electronegativity

Ionization Energy:

Not only is there a correlation between electron configuration and physical properties, but a close correlation also exists between electron configuration (a microscopic property) and chemical behavior (a macroscopic property). As we will see throughout this book, the chemical properties of any atom are determined by the configuration of the atom's valence electrons. The stability of these outermost electrons is reflected directly in the atom's ionization energies. Ionization energy is the minimum energy (in kJ/mol) required to remove an electron from a gaseous atom in its ground state. In other words, ionization energy is the amount of energy in kilojoules needed to strip 1 mole of electrons from 1 mole of gaseous atoms. Gaseous atoms are specified in this definition because an atom in the gas phase is virtually uninfluenced by its neighbors and so there are no intermolecular forces (that is, forces between molecules) to take into account when measuring ionization energy.

The magnitude of ionization energy is a measure of how “tightly” the electron is held in the atom. The higher the ionization energy, the more difficult it is to remove the electron. For a many-electron atom, the amount of energy required to remove the first electron from the atom in its ground state,

$$\text{energy} + X(g) \longrightarrow X^{+}(g) + e^{-}$$ is called the first ionization energy (I1).

In Equation, X represents an atom of any element and e^2 is an electron. The second ionization energy (I2) and the third ionization energy (I3) are shown in the following equations:

$$\text{energy} + X^{+}(g) \longrightarrow X^{2+}(g) + e^{-} \quad \text{second ionization}$$
$$\text{energy} + X^{2+}(g) \longrightarrow X^{3+}(g) + e^{-} \quad \text{third ionization}$$

The pattern continues for the removal of subsequent electrons. When an electron is removed from an atom, the repulsion among the remaining electrons decreases. Because the nuclear charge remains constant, more energy is needed to remove another electron from the positively charged ion.

Thus, ionization energies always increase in the following order: I1 ‹ I2 ‹ I3….

8.5 Electron Affinity

Another property that greatly influences the chemical behavior of atoms is their ability to accept one or more electrons. This property is called ***electron affinity,*** which is *the negative of the energy change that occurs when an electron is accepted by an atom in the gaseous state to form an anion.*

$$X(g) + e^- \longrightarrow X^-(g) \tag{8.4}$$

Consider the process in which a gaseous fluorine atom accepts an electron:

$$F(g) + e^- \longrightarrow F^-(g) \qquad \Delta H = -328 \text{ kJ/mol}$$

The electron affinity of fluorine is therefore assigned a value of +328 kJ/mol. The more positive is the electron affinity of an element, the greater is the affinity of an atom of the element to accept an electron. Another way of viewing electron affinity is to think of it as the energy that must be supplied to remove an electron from the anion. For fluorine, we write

$$F^-(g) \longrightarrow F(g) + e^- \qquad \Delta H = +328 \text{ kJ/mol}$$

Thus, a large positive electron affinity means that the negative ion is very stable (that is, the atom has a great tendency to accept an electron), just as a high ionization energy of an atom means that the electron in the atom is very stable.

Overview of Lanthanides and Actinides

Lanthanides and Actinides are radioactive in nature. They consist of 30 elements in total. They contain radioactive elements such as Uranium. These elements are kept in the lanthanide and actinide series of periodic tables.

What Makes These Elements Special?

Actinides are radioactive elements. Due to their property of radioactivity, they are especially useful in nuclear reactions, and they emit radiation. One Lanthanide (Promethium and all its isotopes) is radioactive, while the others are not. The total number of radioactive elements in Lanthanides and Actinides is 16. If someone asks to **give the general configuration of Lanthanides and Actinides,** you can tell that Actinides have a general configuration of $4f^{n}\,5d^{0}\,6s^{2}$ and that of Lanthanides $5f^{1-14}\,6d^{0-1}\,7s^{2}$

Lanthanides

Lanthanides contain elements from the atomic numbers 57 to 71. These elements are named after the first element in the group called lanthanum, as it exhibits similar chemical properties to all other elements in the group.

It contains 15 elements and is referred to as Ln by chemists. The elements in this group include Lanthanum, Cerium, Praseodymium, Neodymium, Promethium, Samarium, Europium, Gadolinium, Terbium, Dysprosium, Holmium, Erbium, Thulium, Ytterbium, and lutetium.

These 15 elements are also called Rare Earth Metals and are all silvery-white reactive solid metals obtained from the same ore.

57	58	59	60	61	62	63	64
La	Ce	Pr	Nd	Pm	Sm	Eu	Gd
Lanthanum	Cerium	Praseodymium	Neodymium	Promethium	Samarium	Europium	Gadolinium
138.905	140.116	140.908	144.242	144.913	150.36	151.964	157.25
$4f^0 5d^1 6s^2$	$4f^2 5d^0 6s^2$	$4f^3 5d^0 6s^2$	$4f^4 5d^0 6s^2$	$4f^5 5d^0 6s^2$	$4f^6 5d^0 6s^2$	$4f^7 5d^0 6s^2$	$4f^7 5d^1 6s^2$

65	66	67	68	69	70	71
Tb	Dy	Ho	Er	Tm	Yb	Lu
Terbium	Dysprosium	Holmium	Erbium	Thulium	Ytterbium	Lutetium
158.925	162.500	164.930	167.259	168.934	173.055	174.967
$4f^9 5d^0 6s^2$	$4f^{10} 5d^0 6s^2$	$4f^{11} 5d^0 6s^2$	$4f^{12} 5d^0 6s^2$	$4f^{13} 5d^0 6s^2$	$4f^{14} 5d^0 6s^2$	$4f^{14} 5d^1 6s^2$

Lanthanides and their Electronic Configuration

Actinides

These are 15 elements that start with the atomic number 89 and go up to the atomic number 103. They are called Actinides as they all display similar chemical properties to the first element of the group Actinium. The symbol An is used to refer to actinides. This group includes mostly man-made elements other than a few that are naturally occurring. It includes Actinium, Thorium, Protactinium, Uranium, Neptunium, Plutonium, Americium, Curium, Berkelium, Californium, Einsteinium, Fermium, Mendelevium, Nobelium, and Lawrencium.

89 Ac Actinium 227.028	90 Th Thorium 232.038	91 Pa Protactinium 231.036	92 U Uranium 238.029	93 Np Neptunium 237.048	94 Pu Plutonium 244.064	95 Am Americium 243.061	96 Cm Curium 247.070
$5f^0 6d^1 7s^2$	$5f^0 6d^1 7s^2$	$5f^2 6d^1 7s^2$	$5f^3 6d^1 7s^2$	$5f^4 6d^1 7s^2$	$5f^6 6d^0 7s^2$	$5f^7 6d^0 7s^2$	$5f^7 6d^1 7s^2$
97 Bk Berkelium 247.070	98 Cf Californium 251.080	99 Es Einsteinium [254]	100 Fm Fermium 257.095	101 Md Mendelevium 258.1	102 No Nobelium 259.101	103 Lr Lawrencium [262]	
$5f^9 6d^1 7s^2$	$5f^{10} 6d^0 7s^2$	$5f^{11} 6d^0 7s^2$	$5f^{12} 6d^0 7s^2$	$5f^{13} 6d^0 7s^2$	$5f^{14} 6d^0 7s^2$	$5f^{14} 6d^1 7s^2$	

Actinides and their Electronic Configuration

Differences between Lanthanides and Actinides

Differences between Lanthanides and Actinides are listed in the table below.

Serial no.	Lanthanides	Actinides
1	Contain elements with atomic numbers 58 to 71.	Consist of elements from atomic numbers 89 to 103.
2	Non-radioactive except for one element called Promethium.	All elements in this group are radioactive.
3	All naturally occurring elements.	Contain naturally occurring as well as synthetic elements such as plutonium.
4	Common Electronic configuration - $5f^{1-14}\ 6d^{0-1}\ 7s^2$	Common Electronic configuration- $4f^{n}\ 5d^{0}\ 6s^2$
5	Compounds formed by them tend to be less basic.	Compounds formed by them tend to be highly basic in nature.

Electronic Configuration:

Example:

1. Magnesium (Mg) belongs to Group 2A,
Period 3
Atomic # = 12
Electron Configuration = $1s^2 2s^2 2p^6 3s^2$

2. Arsenic (As) belongs to Group 5A, Period 4
Atomic # = 33
Electron Configuration = $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^3$

3. Chlorine (Cl) belongs to Group 7A, Period 3
Atomic # = 17
Electron Configuration = $1s^2 2s^2 2p^6 3s^2 3p^5$

3. Iron (Fe) belongs to Group 8B,
Period 4
Atomic # = 26
Electron Configuration = $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$